

Problem 1

Let $v_1 = (1, -2, 3)$ and $v_2 = (0, 5, -3)$. Verify that the set $\{v_1, v_2\}$ is independent and extend it to a basis of $\mathbf{R}^3$.



Solution 1

Independence amounts to checking that the system of equations generated by $\alpha v_1 + \beta v_2 = \mathbf{0}$ has only the zero solution. This system of equations is the matrix with columns v_1, v_2 acting on $v = (\alpha, \beta)$, so we row-reduce to check.

$$\begin{bmatrix} 1 & 0 \\ -2 & 5 \\ 3 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 \\ 0 & 5 \\ 0 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix},$$

so the columns cannot annihilate each other (which would mean there is only one pivot) via row operations, i.e independent. More simply, if $v_1 = cv_2$ for some c, in the first component we would have $1 = c\ddot{0} = 0$, an impossibility. So, they cannot be multiples of each other and for 2 vectors this means they are independent.

By the plus/minus lemma, suffices to find a vector not in the span of these. Take $v_3 = (0, 0, 1)$ (must have $\alpha = 0$ since first component is zero, and then this then forces $\beta = 0$ since second component is zero, and then this is not possible). Take $v_3 = (0, 0, 1)$. As a check, we can reduce:

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 5 & 0 \\ 3 & -3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

so the only solution is trivial and these are independent.



Problem 2

Let $v_1 = (1, 0, 0, 0)$ and $v_2 = (1, 1, 0, 0)$. Verify that the set $\{v_1, v_2\}$ is linearly independent and extend it to a basis of $\mathbf{R}^4$.



Solution 2

These are clearly not multiples of each other, so these are independent. Simply add $v_3 = (0, 0, 1, 0)$ and $v_4 = (0, 0, 01)$, then an arbitrary element is of the form $(a, b, c, d) = (a - b)v_1 + bv_2 + cv_3 + dv_4$, which gets all of $\mathbf{R}^4$.



Problem 3

Let $v_1 = (1, 0, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (2, 0, 1)$, $v_4 = (0, 0, -1)$. Find a basis of the subspace $\text{span}(v_1, v_2, v_3, v_4)$ of $\mathbf{R}^3$.



Solution 3

We see v_2 is a combination of v_1 and v_4, and that v_3 is a combination of v_1, v_4 are independent. Removing these by plus/minus, we have v_1, v_4 which are not multiples of each other and so indepnedent.

Problem 4

Let $v_1 = (1,1,1,1)$, $v_2 = (2,2,2,0)$, $v_3 = (0,0,0,3)$, $v_4 = (3,3,3,4)$. Find a basis of the subspace $\text{span}\{v_1, v_2, v_3, v_4\}$ of $\mathbf{R}^3$.

Notice $v_4 \in \text{span}(v_1, v_2, v_3)$, so by plus/minus lemma, we can remove v_4 and still span the subspace. Also notice v_2 is a combination of v_1 and v_3, so we can remove this from $\text{span}(v_1, v_2, v_3) = \text{Span}(v_1, v_2, v_3, v_4)$. Now, it is clear that v_1, v_2 are not multiples of each other, so they are independent.

Problem 5

Let $V = \mathbf{R}^{\mathbf{R}}$, the vector space of all functions $f : \mathbf{R} \to \mathbf{R}$. Prove that for every positive integer n, we can find $n+1$ linearly independent functions in V (*hint:* look for polynomials). Explain why it follows that $\mathbf{R}^{\mathbf{R}}$ is infinite-dimensional.

Take the polynomials (as functions) $1, x, ..., x^{n-1}$. These are all independent (as we saw with general polynomials), so this provides n independent vectors. But then, consider the set $\{1, x, ..., x^{n-1}, x^n\}$ which is also independent.

If there exists a finite basis for V, then by this argument, we can find an independent set with more elements. But, this contradicts the theorem that if there is an independent set with m elements in finite-dimensional V, any set with $> m$ elements is dependent. So, V cannot be finite-dimensional.

Problem 6

Determine whether each of the following statements is true or false, and justify your answer:

(a) $\dim\{0\} = 0$.

(b) There is a set of 17 linearly independent vectors in F^{17}.

(c) There is a set of 11 vectors that span F^{17}.

(d) Every linearly independent set of five vectors in F^5 is a basis of F^5.

(e) Every set of five vectors that spans F^5 is a basis of $\mathbf{R}^5$.

(f) There are at least two distinct three-dimensional subspaces of $\mathcal{P}_2(F)$ (the vector space of polynomials of degree at most 2).

Solution 4

(a) By definition, this has basis $\emptyset$, which has size 0, so $\dim\{0\} = 0$.

(b) Yes, take the e_i for $1 \leq i \leq 17$.

(c) No - if there existed such a set, we could reduce it to a basis for F^{17} of size ≤ 11. But, we already know a basis for F_{17} (namely, the e_i) of size 17, and if $s \leq 11$ $s \neq 17$, so no such set can exist.

(d) Yes. We claim every such set spans F^5. For, if it does not, we can extend it to a set of n vectors with $n > 5$ that does span F^5, making it a basis for F^5, so $\dim(F^5) > 5$, impossible since we know $\dim(F^5) = 5$.

(e) This doesn't really make sense?

(f) No - any three-dimensional subspace, by the same argument as in 4, must be all of $P_2(F)$ since $P_2(F)$ is dim 3.